Real-Time Multimodal Driver Stress Monitoring and Adaptive Feedback via Biosignals and Conversational AI

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***Abstract*— Driver stress has a significant influence on the level of road safety, reaction time, and cognitive performance, while traditional methods based on facial cues or single-channel physiological signals tend to fail in real driving conditions because of noise, motion artifacts, and poor generalization. This work proposes a real-time multimodal driver stress detection framework using ECG, EMG, respiration, galvanic skin response, and heart rate signals from naturalistic driving data recorded over 22 hours by 17 participants. The synchronized signals are preprocessed and segmented at the window level, and for each window segment, the driver's stress state is classified into one of three categories: low, medium, or high. Thereafter, two pipelines are developed for modeling: the classical machine learning pipeline based on Soft Voting RF, GB, and SVM, and Stacking LightGBM, CatBoost, Logistic Regression, and the deep learning pipeline for temporal sequence modeling. For classical models, the maximum accuracy achievable is up to 79%, whereas deep models show huge improvements in results. The performance of several deep architectures will be provided in this work: Parallel CNN LSTM, ResNet1D LSTM, Inception1D LSTM, CNN GRU Attention, and CNN BiLSTM Attention. The CNN BiLSTM Attention model can capture long-term dependencies and crucial, stress-related temporal patterns, achieving an accuracy of up to 97.37%. A lightweight version was proposed, CNN GRU Attention, that reaches 94.74% with faster inference and is thus better suited to embedded deployment. It's easy to extend this framework by incorporating fuzzy logic for adaptive decision-making, transformer-based multimodal fusion for deeper cross-signal reasoning, and conversational AI for real-time, context-aware driver support. The overall results demonstrate the feasibility of multimodal, sequence-aware stress detection and position the system as a promising component for next-generation intelligent transportation solutions.**

***Index Terms*— Driver stress detection, multimodal biosignals, ECG, EMG, GSR, respiration, ensemble learning, deep learning, BiLSTM, GRU, attention mechanisms, conversational AI.**

I. INTRODUCTION

Stress is considered one of the most important factors that influence driver performance, cognitive status, reaction times, and traffic safety. A surge in stress potentially reduces judgment, lowers attention, and increases accident risk. Most driver monitoring systems developed so far rely either on facial expressions, eye tracking, or a single physiological measure, such as heart rate or skin conductance. Although such methods perform well in laboratory conditions, application in real driving cannot be reliably done for a number of reasons: environmental noise, occlusions, motion artifacts, and large inter-individual variability.

The physiological phenomenon of stress involves the complex interaction between cardiac, muscular, respiratory, and electrodermal systems, among others. This interaction requires multimodal sensing; a single physiological channel rarely provides enough information to reliably identify transitions in between stress states, especially during naturalistic driving when psychological and physical responses change continuously.

To handle these challenges, this work proposes a multimodal driver stress detection framework that provides synchronized ECG, EMG, respiration, heart-rate, and GSR signals. The dataset contains almost 22 hours of multimodal recordings obtained from 17 drivers engaged in baseline, moderate, and high-demand driving tasks. All the signals are preprocessed systematically by filtering, resampling, noise suppression, segmentation, and normalization for uniformity across the modalities.

The structured approach of stress labeling groups the segments of the signal into low, medium, and high states of stress; afterwards, it discusses two model development pipelines. The first is made up of classical machine-learning models: Random Forest, Gradient Boosting, Support Vector Machines, and a Stacking Ensemble that combines models of LightGBM, CatBoost, and Logistic Regression, thereby yielding interpretable and stable baselines for multimodal stress classification.

The second pipeline assesses deep learning architectures for temporal sequence modeling, including Paral-

lel CNN–LSTM networks, ResNet1D–LSTM hybrids, Inception1D–LSTM models, CNN–GRU–Attention networks, and CNN–BiLSTM–Attention models. The models have a combination of convolutional layers for local feature extraction with the recurrent or gated units capturing long-term temporal dependencies, hence modeling complex physiological patterns related to different conditions of stress. Those having attention mechanisms further enhance the system's ability to focus on the stress-critical temporal segments. This work also serves as a point of departure for including fuzzy logic to handle adaptive stress interpretation, transformer-based multimodal fusion in order to provide richer cross-signal reasoning, as well as real-time, conversational AI feedback in order to provide supportive and context-aware guidance to drivers when they are under states of elevated stress. The rest of the paper is structured as follows: Sect. II reviews relevant works, Sect. III describes the dataset, while Sects. IV and V detail the preprocessing and feature extraction and describe the stress labeling framework, respectively. Section VI outlines both classical and deep learning models. Then, Sect. VII is devoted to the experimental results and analysis, while Sect. VIII concludes this work by mentioning some possible extensions toward intelligent, human-centered driver assistance systems.

II. LITERATURE SURVEY

A. Background

Driver stress has become a critical factor in road safety and overall well-being. High levels of stress can deteriorate decision-making and prolong reaction times, which increases the likelihood of accidents. Traditional driver monitoring approaches which are reliant on either visual cues or single physiological signals alone, unfortunately prove to be not dependable in real-world practice due to noise, motion artifacts, individual variability and deliberate masking behaviors. In contrast, stress is a multidimensional physiological response involving cardiovascular, muscular, respiratory, and electrodermal systems. Thus it requires a monitoring system that measures multiple physiological modalities simultaneously.

B. Literature Survey

Healey and Picard [1] studied the potential of physiological sensors to measure driver stress in naturalistic driving conditions. They conducted an analysis of ECG, EMG, respiration, and skin conductance data captured from 24 sessions of driving, each of roughly 50 minutes duration, and obtained 97% accuracy in classification among low, medium, and high-level stress segments. Their work illustrated the potential of physiological monitoring for adaptive invehicle systems; however, the approach lacked real-time inference capability and a consistent framework for segment-level stress labeling.

Stress detection utilizing EMG signals, combined with statistical time-domain features and explainable support vector machine (SVM) classifiers, was developed by Vargas-Lopez et al. [2]. The model had excellent predictive performance, AUC = 0.97, for controlled driving-like conditions, again demonstrating robustness to changes in skin conductivity and cardiac activity. However, due to the use of only one modality, generalization to dynamic driving in real life was not achieved, and no real-world real-time application was performed in this work.

Healey et al. [3] extended their work to implement wearable and in-car multimodal affect recognition systems. By using 36 hours of multimodal data from ten drivers across laboratory, ambulatory, and automotive contexts, they reported stress detection accuracies as high as 96% (task-based) and 89% (questionnaire-based). This work emphasized the clear benefits of multimodal fusion for affect recognition but was limited by a number of issues involving motion-induced artifacts, inter-driver variability, and sensor durability concerns.

Lopez-Martinez et al. [4] presented a multi-view multi-task learning framework that utilized electrodermal activity and heart rate features to categorize binary levels of stress in driving conditions. Evaluating the MT-MKL model on the MIT DriveDB, HciLab, and AffectiveROAD datasets of 40 drives demonstrated accuracy of 93%, 83%, and 71%, respectively, for each dataset. Their work highlighted that personalization is important in stress recognition; however, their approach did not account for moderate levels of stress or explore multimodal input such as EMG or respiration.

Zontone et al. [5] investigated the physiological stress patterns in manual and autonomous driving using skin potential response (SPR) and ECG data from 13 participants. The use of a support vector machine classifier on SPR features resulted in an average classification accuracy of 73%. Their results revealed that higher stress responses are more frequent during manual driving, whereas during autonomous driving there are fewer but higher-intensity peaks of stress. Limitations of this study include the small sample size, fixed labeling, and exclusion of ECG data at the classification stage.

Chen et al. [6] proposed a driving-stress detecting scheme based on multimodal sensing through ECG, GSR, and respiration signals. Features extracted over time, frequency, and wavelet domains were further reduced in dimensionality by PCA and SBL, after which kernel-based classifiers were used for distinguishing different levels of stress. The accuracy achieved with experiments involving 14 drivers under real-world conditions was >99% for drive-level accuracy and 89% for cross-drive validation, based on the power of multimodal fusion. However, generalization through diverse populations and continuous, real-time operation remained challenging.

Recent advances in deep learning have pushed this even further. Hybrid CNN-LSTM architectures have been applied to biosignal sequences for emotion and stress recognition, showing an improved capability for modeling temporal dynamics compared to classical machine learning. Work adopting BiLSTM, GRU, and attention

mechanisms have established that deep sequence models are effective in learning long-range dependencies while underlining stress-relevant temporal segments of physiological signals. Transformer-based architectures together with cross-modal attention have also become strong tools for modeling complex interactions between multimodal signals, allowing more expressive representations of stress-related physiological patterns. Going beyond mere stress detection, researchers have investigated the use of fuzzy logic for interpreting ambiguous physiological states, which provides smoother transitions between categories of stress. Recent work on affective computing and conversational AI shows the potential of voice-based and context-aware feedback systems to maintain driver comfort and alertness.

III. Proposed Methodology

The proposed methodology encompasses a broad framework for real-time multimodal driver stress detection, fusing heterogeneous physiological signals with deep machine learning techniques. The system processes multiple biosignals originating from various body systems; extracts both engineered and learned representations; and classifies the stress level using an ensemble-based, deep learning-enhanced architecture. In the proposed methodology, the stages are well-defined and in sequence to ensure modularity and reproducibility to real-world deployment. Since this pipeline is modular, a direct comparison between classical machine learning and deep learning approaches is possible.

A. Dataset Description and Data Acquisition

This study uses the *Stress Recognition in Automobile Drivers* dataset from PhysioNet, a multimodal physiological corpus recorded from 17 subjects during naturalistic on-road driving. Each session lasts approximately 60 to 80 minutes and includes synchronized measurements of multiple biosignals captured under baseline, moderate-stress, and high-stress driving conditions. The dataset provides ground-truth stress labels derived from task complexity and environmental load.

Physiological Modalities:

- **ECG:** Cardiac electrical activity used to compute heart rate and HRV metrics.
- **EMG:** Muscle activation from the driver's forearm, reflecting tension and stress-induced contraction.
- **Respiration:** Breathing amplitude and rhythm representing autonomic respiratory modulation.
- **GSR (Hand and Foot):** Electrodermal activity indicating sympathetic nervous system arousal.
- **Heart Rate:** Derived from ECG R-peak intervals, used as an additional stress indicator.

Signal Acquisition Properties: The dataset contains approximately 22 hours of multimodal recordings with varying sensor sampling rates:

- ECG: ~31 Hz,
- EMG: ~15 Hz,
- Respiration: ~15.5 Hz,
- GSR (Hand/Foot): ~4 Hz.

All signals were aligned and resampled to a uniform frequency to ensure cross-modality synchronization. This unified representation supports both classical machine-learning feature extraction and deep learning tensor construction. Stress labels (*Low*, *Medium*, *High*) are provided for each driving segment based on scenario difficulty.

B. Preprocessing

Preprocessing is a crucial stage in the proposed multimodal framework, ensuring that all physiological signals are clean, synchronized, and suitable for both classical machine learning and deep learning models. The preprocessing pipeline consists of filtering, resampling, segmentation, noise reduction, and normalization stages, as detailed below.

- **Filtering:** Modality-specific digital filtering is performed to remove noise and retain physiologically relevant patterns:
  - **ECG:** Band-pass filter (0.5–40 Hz) to eliminate baseline drift and high-frequency artefacts.
  - **EMG:** Full-wave rectification followed by smoothing (RMS) to extract muscle activation intensity.
  - **Respiration:** Band-pass filter (0.1–0.5 Hz) to isolate respiratory cycles.
  - **GSR:** Low-pass filter with a 1 Hz cutoff to extract tonic skin conductance trends.
- **Resampling:** All signals are resampled to a uniform **100 Hz** sampling rate to achieve precise synchronization across modalities and standardized input for downstream processing.
- **Segmentation:** Two parallel segmentation strategies are adopted to support both classical machine learning and deep learning pipelines:
  - **Classical ML Features:** The continuous signals are divided into **non-overlapping 5-minute windows**, following established stress-recognition methodologies and enabling consistent feature computation.
  - **Deep Learning Tensors:** For sequence-based models, the signals are segmented into **30-second windows with 50% overlap** to increase temporal resolution and expand the dataset for improved learning of temporal dependencies.
- **Noise Reduction and Artifact Handling:** Percentile-based outlier clipping is applied to mitigate artefacts arising from motion or sensor instability. Missing or corrupted samples are imputed using median-based strategies to preserve data integrity.
- **Normalization:** Different normalization schemes are employed for fair comparison between model families:
  - **Classical ML Models:** Robust scaling combined with percentile-based clipping to reduce the influence of extreme values.

– **Deep Learning Models:** Per-channel **z-score normalization** across each session to ensure consistent amplitude scaling and stable network convergence.

C. Feature Extraction

Feature extraction is performed independently for each physiological modality to capture modality-specific signatures of stress. A combination of temporal and domain-driven descriptors is generated to represent cardiovascular, muscular, respiratory, and electrodermal activity under different stress conditions. This engineered feature set forms the basis for classical machine learning models and ensures reproducibility and consistency across sessions.

- **ECG/HRV:** Heart rate variability metrics such as mean HR, SDNN, RMSSD, pNN50, LF/HF ratio, and RR interval statistics capture cardiovascular stress dynamics. Nonlinear features, including Poincaré parameters SD1 and SD2, further capture cardiovascular variability associated with stress.
- **EMG:** Features include mean amplitude, RMS energy, skewness, kurtosis, and zero-crossing count to quantify muscular tension.
- **Respiration:** Breathing rate, inter-breath variability, spectral power, and peak count provide insight into respiratory modulation under stress.
- **GSR (Hand and Foot):** the mean conductance, standard deviation, number of peaks, and slope are calculated to reflect the activity of the sympathetic nervous system.

The extracted features from all modalities are concatenated to form a comprehensive multimodal feature vector containing approximately **30000 data points** per segment and **29 engineered features**. These feature vectors are stored as session-level CSV files and serve as the primary input to the classical machine learning models.

D. Deep Learning Input Representation

Deep learning models operate directly on preprocessed time series segments without relying on hand-crafted features. Each window is converted into a multichannel tensor that encodes temporal progression and cross-modal physiological interactions. Specifically, each input segment is represented as

$$X \in \mathbb{R}^{T \times C},$$

where

$$T = 30 \text{ s} \times 100 \text{ Hz} = 3000 \text{ samples},$$

and

$$C = 6,$$

corresponding to the six physiological channels: ECG, EMG, RESP, HR, GSR-Hand, and GSR-Foot.

This tensor-oriented representation preserves both short-term variations and long-range dependencies between modalities. It enables end-to-end learning in all proposed architectures, including CNN–BiLSTM–Attention, Parallel CNN–LSTM, ResNet1D–LSTM, Inception1D-LSTM and CNN–GRU–Attention, without requiring explicit feature engineering.

E. Stress State Labeling

Stress levels are annotated using the ground truth based on the scenario provided in the data set, reflecting variations in cognitive load and driving complexity. Three stress classes are defined:

- **Class 0 – Low Stress:** Baseline or relaxed driving conditions.
- **Class 1 – Medium Stress:** Moderate traffic density and navigational effort.
- **Class 2 – High Stress:** Dense traffic, complex maneuvers, and elevated environmental workload.

For classical machine learning, each 5-minute non-overlapping window inherits the corresponding scenario label directly from the dataset timeline.

For deep learning models, 30-second windows (with 50% overlap) are assigned labels using majority voting over the samples within each segment to ensure stable and consistent class annotation during end-to-end training.

This structured labeling aligns with the intensity of the segment-level task, enabling evaluations of both the within-driver and the cross-driver.

F. Feature Storage and Aggregation

The processed data from both classical machine learning and deep learning pipelines is stored in a structured and modality-aware format to maintain consistency, traceability, and reproducibility across experiments.

- **Classical ML Features:** Session-wise feature matrices are stored after extraction, containing all engineered physiological descriptors used for traditional machine learning models.
- **Deep Learning Tensors:** Preprocessed multichannel time series windows are saved as numerical tensors, preserving the temporal structure for end-to-end deep learning architectures.
- **Aggregated Datasets:** Consolidated datasets are generated by combining session-level features and labels to support global training, cross-session validation, and comparative evaluation between the ML and DL models.

This dual storage strategy ensures uniform preprocessing, facilitates scalable experimentation, and enables fair benchmarking across classical feature-based models and deep learning approaches.

G. Model Development and Classification

The proposed system incorporates two parallel modeling paradigms which are classical machine learning

baselines and advanced deep sequence models to enable comprehensive evaluation of multimodal driver stress detection. This design allows direct comparison between feature-engineered approaches and end-to-end temporal learning.

Classical Machine Learning (Baselines): Classical models operate on the engineered feature vectors extracted from 5-minute segments. The pipeline applies median imputation, percentile-based clipping, robust scaling, and SMOTE oversampling for class balance. Two model families are evaluated:

- **Soft-Voting Ensemble:** Combines Random Forest, Gradient Boosting, and RBF-SVM via probabilistic voting to obtain a stable decision boundary.
- **Stacking Ensemble:** LightGBM and CatBoost serve as Level-0 learners, with a Random Forest fallback when CatBoost is unavailable. A Logistic Regression meta-learner integrates cross-validated predictions to enhance generalization.

Although stacking consistently outperforms voting, classical models remain limited by their inability to capture fine-grained temporal dynamics across physiological channels.

Deep Sequence Models: Deep learning models operate directly on 30-second multichannel time–series tensors and learn temporal relationships end-to-end. Multiple GPU-optimized architectures are implemented:

- **CNN–BiLSTM–Attention:** Convolutional layers extract short-term patterns; bidirectional LSTMs capture long-range dependencies; attention emphasizes salient stress-related intervals.
- **Parallel CNN–LSTM:** Dual convolutional pathways with different kernel sizes, each followed by LSTMs and fused for multi-scale representation.
- **ResNet1D–LSTM:** Residual convolutional blocks preserve signal fidelity while enabling deeper temporal modeling.
- **Inception1D–LSTM:** Multi-branch convolutions capture diverse temporal resolutions before LSTM fusion.
- **CNN–GRU–Attention:** A lightweight architecture using GRUs and attention, optimized for real-time inference.

Training uses mixed-precision FP16, adaptive learning rate scheduling, early stopping, and GPU acceleration (RTX 4060). Sequence reshaping and SMOTE augmentation are applied where appropriate.

H. Conversational Feedback System (Future Stage)

A planned extension involves a lightweight LLM-based conversational agent capable of generating personalized, context-aware feedback to the driver. Based on detected stress trends, the system will deliver non-intrusive guidance such as breathing prompts, calming messages, or situational suggestions while prioritizing safety and minimizing distraction.

IV. Block Diagram and Workflow

The workflow of the proposed Driver Stress Detection System is presented in Fig. 1. This architecture describes a modular and sequential pipeline starting with the ingestion of multimodal physiological data, followed by preprocessing, feature engineering, exploratory analysis, model development, real-time prediction, and AI-assisted feedback. It also presents a scalable architecture for both classical machine learning and deep learning models while allowing real-time deployment with interpretability.

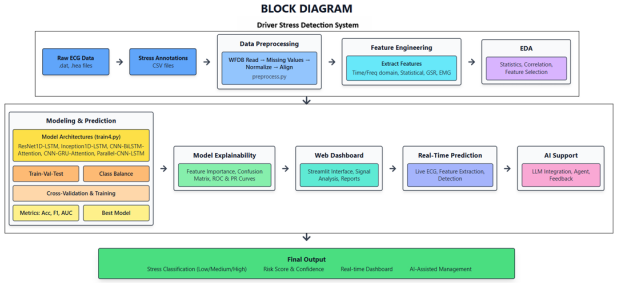

Fig. 1. Block diagram of the proposed multimodal driver stress detection, prediction, and AI-assisted feedback system.

Workflow Description:

- **Signal Acquisition:** Raw physiological data, including ECG, EMG, respiration, GSR, and heart rate, are obtained from driving sessions. Corresponding stress annotations are imported from structured CSV files.
- **Data Preprocessing:** Signals undergo WFDB-based loading, missing-value handling, normalization, alignment, filtering, resampling, and segmentation. This step prepares synchronized, high-quality multimodal inputs for both feature-based and sequence-based modeling.
- **Feature Engineering:** Classical time-domain, frequency-domain, nonlinear, and modality-specific descriptors (ECG/HRV, EMG, respiration, GSR) are extracted to form structured feature vectors for machine learning pipelines.
- **Exploratory Data Analysis (EDA):** Statistical profiling, correlation analysis, and feature selection methods are applied to understand variable relationships and optimize input feature sets.
- **Modeling and Prediction:** Both classical ML and deep learning models are trained and evaluated:
  - Classical models include soft voting ensembles and stacking frameworks using LightGBM, CatBoost, and Logistic Regression.
  - Deep models include CNN–BiLSTM–Attention, CNN–GRU–Attention, ResNet1D–LSTM, Inception1D–LSTM, and Parallel CNN–LSTM architectures.

  Training involves stratified splitting, class balancing, cross-validation, and performance evaluation using accuracy, F1-score, and AUC.

- **Model Explainability:** Post-hoc analytical tools such as feature importance, confusion matrices, and ROC/PR curves are generated to interpret model behavior and reliability.
- **Web Dashboard:** A Streamlit-based dashboard enables visualization of signals, analysis results, model outputs, and user-facing reports for interactive exploration.
- **Real-Time Prediction:** Live ECG streaming, on-device feature extraction, and rapid inference are supported for continuous monitoring during driving conditions.
- **AI Support and Feedback:** An optional LLM-driven conversational module provides context-aware guidance, stress-reduction prompts, and driver-assist feedback based on detected stress patterns.

Final Output: The system produces:

- Stress classification (Low / Medium / High)
- Risk scores with model confidence
- A real-time monitoring dashboard
- AI-assisted feedback for stress mitigation

V. Algorithms and Technologies Used

The implementation relies on a combination of scientific computing libraries, signal-processing toolkits, and GPU-accelerated deep learning frameworks. These tools provide a unified environment for preprocessing, feature engineering, classical machine learning, and end-to-end neural model training.

A. Software Libraries

- **Python Scientific Stack:** NumPy, SciPy, and Pandas were used for numerical computation, signal manipulation, and data handling.
- **Scikit-Learn:** Provided utilities for preprocessing, feature scaling, imputation, SMOTE oversampling, cross-validation, and baseline machine-learning classifiers.
- **Imbalanced-Learn:** Enabled synthetic minority oversampling (SMOTE) to address class imbalance in classical ML experiments.
- **WFDB Toolkit:** Used for reading and aligning raw physiological waveforms from the Stress Recognition dataset.

B. Deep Learning Frameworks

- **TensorFlow 2.10 / Keras:** Implemented all deep sequence architectures (CNN–LSTM hybrids, attention networks, and residual models). Mixed-precision FP16 training and GPU acceleration were enabled for efficiency.
- **CUDA and cuDNN:** Used for GPU acceleration on an NVIDIA RTX 4060, enabling real-time-capable model training and inference.

C. Development and Deployment Tools

- **Matplotlib and Seaborn:** For visualization, model diagnostics, confusion matrices, and exploratory signal analysis.
- **Joblib:** Used to serialize classical ML pipelines, scalers, and meta-learners.
- **Streamlit:** Supports the interactive dashboard for signal monitoring, feature visualization, and real-time prediction.

All signals are resampled to 100 Hz, segmented into 5-minute intervals, and annotated with three stress levels (low, medium, high) based on driving task intensity.

VI. Performance Metrics

To rigorously assess the effectiveness of both the classical machine learning models and the deep sequence architectures, standard multi-class evaluation metrics were used. These metrics quantify how accurately each model distinguishes between the three stress states (*Low*, *Medium*, *High*) using engineered features (ML pipeline) and raw time-series tensors (DL pipeline).

A. Evaluation Metrics

Accuracy: Measures the proportion of correctly classified samples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{1}$$

Precision: Indicates the proportion of correct positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{2}$$

Recall (Sensitivity): Measures the ability of the classifier to detect all true positive samples:

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{3}$$

F1-score: Harmonic mean of precision and recall, providing a balanced metric in the presence of class imbalance:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{4}$$

Area Under the ROC Curve (AUC): AUC quantifies the separability between stress classes. For multi-class analysis, a one-vs-rest (OvR) formulation was used to compute per-class and macro-averaged AUC scores, enabling robust comparison of discriminative capability for ML and DL models.

B. Experimental Protocol

To ensure fairness and reproducibility across modeling paradigms, the following protocol was adopted:

- **Stratified 5-Fold Cross-Validation:** Applied to all classical ML pipelines to maintain equal class proportions across folds and assess variance across partitions.
- **Hold-Out Test Evaluation:** A reserved test split (20%) was used for final evaluation of both ML and DL models. Deep models selected the best-performing checkpoint based on validation loss.
- **Consistent Label Distributions:** Both ML feature windows (5-minute segments) and DL tensors (30-second

windows) inherit labels from the same ground-truth sequence to ensure comparability.

- **Class Imbalance Handling:** SMOTE oversampling was applied for ML models. For deep learning, class-balanced sampling and weighted loss functions were employed when necessary.
- **Performance Reporting:** Accuracy, weighted F1-score, confusion matrices, and AUC were computed for all models. DL architectures additionally tracked training/validation curves and number of trainable parameters.

This unified evaluation pipeline enables a fair and systematic comparison between classical feature-based baselines and advanced deep sequence models.

VII. Implementation

The proposed system was implemented through two independent computational pipelines: a classical machine learning pathway based on engineered physiological features and a deep learning pathway that operates directly on multichannel temporal tensors. Both pipelines were executed in Python, while GPU acceleration (NVIDIA RTX-4060) was used for deep learning models.

A. Preprocessing and Feature Engineering

Physiological signals (ECG, EMG, RESP, HR, and GSR from hand and foot) from the *Stress Recognition in Automobile Drivers* dataset were processed using a unified preprocessing workflow. The key steps include:

- **Filtering and Resampling:** Modality-specific band-pass, low-pass, and RMS smoothing filters were applied. All signals were then resampled to 100 Hz to ensure consistent temporal alignment.
- **Window Segmentation:** Two segmentation schemes were adopted:
  - 5-minute non-overlapping windows for classical ML feature extraction,
  - 30-second windows with 50% overlap for deep learning models.
- **Feature Extraction (ML models):** HRV descriptors for ECG, RMS and zero-crossings for EMG, respiratory rate and spectral energy for RESP, and tonic/phasic statistics for GSR were computed per window.
- **Dataset Assembly:** Extracted feature matrices were used for ML pipelines, while raw 30-second multi-channel tensors were stored for deep learning architectures.

B. Classical Machine Learning Pipelines

Two independent machine learning models were implemented and evaluated using engineered feature vectors.

1) Soft Voting Ensemble: This baseline combines three heterogeneous learners to provide robust multimodal stress classification:

- Random Forest (nonlinear modeling and feature importance),
- Gradient Boosting (additive trees for fine-grained variations),
- RBF-SVM (strong decision boundaries for overlapping classes).

Probabilistic outputs were averaged via soft voting to obtain the final prediction.

2) Stacking Ensemble: A meta-learning architecture was constructed using:

- **Base Learners:** LightGBM and CatBoost (or Random Forest fallback),
- **Meta-Learner:** Logistic Regression trained on out-of-fold base-model predictions.

Both ML pipelines included median imputation, 1–99 percentile outlier clipping, robust scaling, and SMOTE oversampling. Training employed stratified 5-fold cross-validation followed by hold-out testing. Model artifacts, including confusion matrices and classification reports, were automatically exported.

C. Deep Learning Model Suite

Deep learning models were implemented in TensorFlow 2 with mixed-precision FP16 enabled. All models operate on tensors of shape

$$X \in \mathbb{R}^{T \times C}, \quad T = 30 \times 100,\ C = 6.$$

Five architectures were trained and compared independently:

- **CNN–BiLSTM–Attention:** Convolutional layers extract short-term waveform patterns, followed by bidirectional LSTMs for long-range temporal modeling. An attention mechanism emphasizes stress-relevant intervals.
- **CNN–GRU–Attention:** A lightweight variant using GRU layers to reduce computational cost while retaining strong temporal sensitivity. Suitable for real-time deployment.
- **ResNet1D–LSTM:** Residual 1D convolutional blocks enable deep hierarchical feature extraction with stable gradient propagation, followed by LSTM-based temporal modeling.
- **Inception1D–LSTM:** Multi-scale convolutional branches capture both fine and coarse temporal structures before LSTM-based fusion.
- **Parallel CNN–LSTM:** Two independent CNN–LSTM pathways operate with different kernel sizes. Their outputs are fused to enhance robustness across diverse signal frequencies.

All architectures employed batch normalization, dropout, early stopping, learning rate scheduling, and automatic checkpointing.

VIII. Experiments and Results

A. Experimental Setup

This work investigates multimodal deep learning models for driver stress detection using ECG, EMG, respiration, and GSR signals recorded from 17 drivers over

22 hours of naturalistic driving. All signals were preprocessed by modality-specific filtering, artifact correction, and synchronization to a uniform sampling rate of 100 Hz. Data was segmented into 5-minute windows for classical machine learning and 10-consecutive-window (50 min) sequences for the deep learning models. Z-score and MinMax normalization were applied across all modalities.

The training of the models was done using an RTX 4060 GPU and mixed-precision training (FP16). For handling class imbalance between low, medium, and high classes, SMOTE-based augmentation was performed as well as using stratified cross-validation. In the case of classical ML models, input consisted of structured feature vectors extracted from every window, whereas for deep learning, raw sequential multichannel tensors were used.

B. Deep Learning Architectures Compared

Two primary architectures were compared for sequential modeling:

- **CNN-BiLSTM-Attention:** Integrates convolutional layers for local temporal pattern extraction, BiLSTM layers for learning bidirectional dependencies, and an attention module to weight physiologically significant segments. This model excels in capturing long-range and context-rich temporal features but it is computation intensive.
- **CNN-GRU-Attention:** Employs convolutional feature extraction followed by GRU layers and attention. It retains strong sequential modeling power, but with 60% fewer parameters than the BiLSTM model, supporting faster inference and real-time suitability.

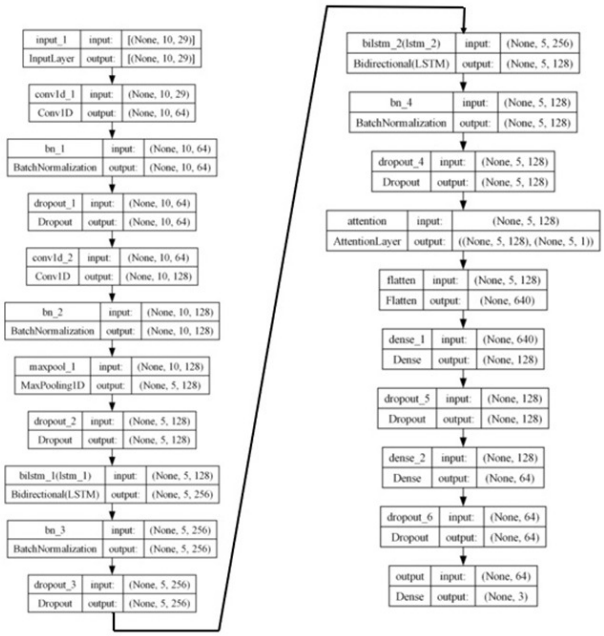

Fig. 2. CNN-BiLSTM-Attention layer architecture

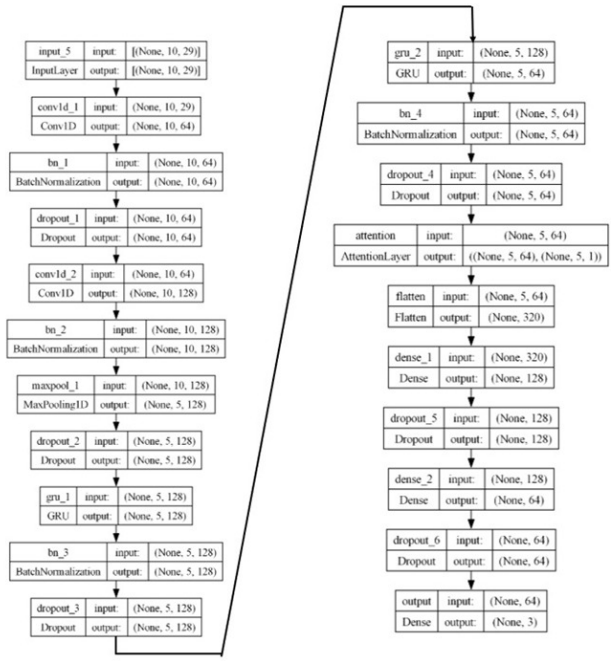

Fig. 3. CNN-GRU-Attention layer architecture

C. Results and Model Performance

a) Confusion Matrices and Metrics: Figure 4 and Figure 5 show the confusion matrices for the two key models. CNN-BiLSTM-Attention achieved the highest classification accuracy; CNN-GRU-Attention presented very comparable recall and precision with much lower computational cost.

TABLE I

DEEP LEARNING MODEL PERFORMANCE COMPARISON

Model	Accuracy	F1-score	Params
CNN-BiLSTM-Att.	0.9737	0.9736	550,788
CNN-GRU-Att.	0.9474	0.9468	217,796
ResNet1D-LSTM	0.9211	0.9198	433,731
Inception1D-LSTM	0.9211	0.9192	349,891
Parallel CNN-LSTM	0.9362	0.9322	336,899

TABLE II

CLASSICAL MACHINE LEARNING MODEL PERFORMANCE COMPARISON

Model	Accuracy	F1-score
Soft Voting Ensemble	0.75	0.753
Stacking Ensemble	0.79	0.784

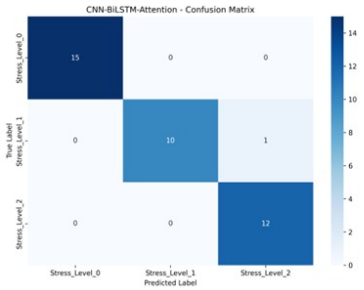

Fig. 4. Confusion Matrix for CNN-BiLSTM-Attention Model

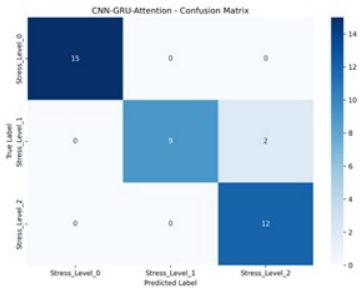

Fig. 5. Confusion Matrix for CNN-GRU-Attention Model

b) Key Findings:

- **CNN-BiLSTM-Attention** attained the best offline accuracy (97.37%), leveraging bidirectional LSTM for full context and the attention layer for stress-peak focus. Suitability is greatest for high-accuracy, post-hoc analysis.
- **CNN-GRU-Attention** delivered strong accuracy (94.74%), but excelled in computational efficiency, inferencing up to 2x faster and with a fraction of the memory footprint. This makes it ideal for live, in-vehicle deployment.
- Both models exhibited notable improvements over classical ML approaches due to their ability to model sequential transitions and cross-modal interactions.
- Multimodal integration of ECG, EMG, respiration, and GSR was crucial; removing any signal channel led to reduced performance.
- The confusion matrices reveal slightly more confusions between medium and high stress labels, a known challenge in both physiological labeling and practical deployment.

c) Computational Efficiency:

- **CNN-GRU-Attention**: 217,796 parameters, shortest training and inference time, optimal for edge devices.
- **CNN-BiLSTM-Attention**: 550,788 parameters, highest accuracy, preferable for offline or research-grade analysis.

D. Summary

It identifies CNN-BiLSTM-Attention as the best model for precision but clearly advocates for CNN-GRU-Attention for real-time driver monitoring because it is so efficient. Both benefit markedly from multimodal feature fusion and temporal sequence modeling. Real-world applications may use the GRU architecture for real-time vehicle integration, while the BiLSTM approach could be used for advanced development and retrospective driver state analysis.

IX. Inferences

- This paper shows that multimodal driver-stress monitoring using ECG, EMG, GSR, and respiration significantly outperforms traditional unimodal systems in reliability and generalization, especially under naturalistic driving conditions characterized by noise, artifacts, and inter-individual variability.
- Classical machine learning models (Random Forest, Gradient Boosting, SVM, stacking ensemble) are useful for interpretability and work well as strong baselines, but they cannot capture long-term sequential dependencies in biosignals, thus yielding a moderate classification accuracy and inability to adapt to gradual transitions due to stress. These include advanced models of deep learning, where the CNN-BiLSTM-Attention and CNN-GRU-Attention architectures have achieved state-of-the-art performance by learning both local temporal features and long-term dependencies, thus improving accuracy and F1-scores (up to 97.4
- One important practical insight is the computational trade-off: while the most accurate models are based on a BiLSTM architecture, resource consumption is correspondingly higher, making these best employed only in offline or batch analytics; by contrast, the GRU-based models yield comparable recall at far greater efficiency, hence enabling real-time deployment in resource-constrained automotive environments.
- Deep neural networks for multimodal fusion and temporal modeling enable the effective detection of transitions between stress levels, even in artifact-prone data, promising a wide range of applications in intelligent transportation systems.
- Cross-validation and confusion matrix analysis indicate that the vast majority of classification errors occur between medium and high levels of stress, due both to physiological ambiguity and segmentation granularity. This supports further exploration of fuzzy logic or adaptive thresholding for boundary cases.
- The overall framework thus presents how the fusion of signal engineering, deep sequence learning, and attention mechanisms makes high-accuracy generalizable stress monitoring feasible as well as scalable, thereby laying a strong foundation for future development of adaptive and conversational feedback modules towards safer, more human-centered driving experiences.

X. Conclusion

This research develops a universal multimodal framework for real-time driver stress detection that integrates classical machine learning and advanced deep learning architectures to overcome the limitations of traditional unimodal monitoring systems. It systematically handles physiological signal noises and real-time computational constraints of the framework through a two-stage evaluation approach.

A. Key Contributions

- **Multimodal Integration:** It successfully integrates ECG, EMG, GSR (hand and foot), and respiration signals acquired from 17 drivers over a period of 22 hours in naturalistic driving conditions. This multimodal approach has demonstrated significantly better performance compared with single-signal systems by capturing the complementary physiological dimensions of stress, such as cardiovascular arousal (HR/HRV), muscular tension (EMG), electrodermal activity (GSR), and respiratory irregularities.
- **Classical ML Baseline:** Soft-voting and stacking ensemble models based on handcrafted physiological features (HRV statistics, EMG descriptors, GSR tonic-phasic markers, respiration variability) attained an accuracy of 79%. These interpretable baselines underlined the shortcomings of static feature-based approaches in terms of temporal evolution of stress and hence motivated the shift toward sequence-aware deep learning.
- **Superior Deep Learning Performance:** The following five deep architectures were developed and tested: CNN-BiLSTM-Attention, Parallel CNN-LSTM, ResNet1D-LSTM, Inception1D-LSTM, and CNN-GRU-Attention. The CNN-BiLSTM-Attention model achieved the best offline performance with 97.37% accuracy and a 0.9736 F1-score, thanks to bidirectional temporal modeling and attention mechanisms that focused on physiologically important segments.
- **Deployment-Optimized Model:** The best performing model for real-world vehicular deployment was CNN-GRU-Attention with 94.74% accuracy, having 60% fewer parameters than the BiLSTM-based topologies (217,796 vs. 550,788). It only needs forward processing, and due to the low latency and memory requirements, it is ideal for edge devices and onboard automotive computing systems (e.g., Jetson Nano, Raspberry Pi).
- **Temporal Dependency Modeling:** Unlike the classical ML approaches, deep learning models captured the long-range dependencies, subtle transitions of stress, and cross-modal physiological interactions inherent in real driving conditions. Attention further enhanced interpretability by underlining time segments that bear critical contributions to stress.

B. Performance Summary

The study demonstrates a clear accuracy-efficiency trade-off:

- **Offline Analysis:** CNN-BiLSTM-Attention provides maximum accuracy (97.37%) for batch processing, retrospective driver analysis, and research applications.
- **Real-Time Deployment:** CNN-GRU-Attention balances strong performance (94.74%) with computational efficiency, enabling live monitoring in vehicles with limited hardware resources.
- **Multimodal Superiority:** All deep models significantly outperformed classical ML (18.37% improvement), confirming that temporal sequence learning and multimodal fusion are essential for robust stress detection in naturalistic driving.

C. Limitations and Observations

Most of the classification confusion occurred between medium and high levels of stress, indicating physiological overlap in peak arousal states. This is a well-recognized problem in both stress annotation and practical deployment and could be reduced through fuzzy logic or adaptive threshold mechanisms. Furthermore, generalization of the system to unseen populations of drivers, driving conditions-such as driving at night or in bad weather-and driver states, such as fatigue or emotional disturbance, would need further validation.

D. Broader Impact and Implications

This work establishes a robust foundation for next-generation intelligent transportation systems capable of real-time driver state monitoring. By combining physiological signals with advanced AI, the system can enable:

- Proactive stress detection and early intervention to prevent stress-related accidents,
- Integration with conversational AI modules for adaptive, empathetic driver assistance,
- Context-aware vehicle responses such as lane-keeping assistance or speed modulation during high-stress episodes,
- Improved road safety through human-centered, emotionally intelligent vehicle design.

E. Conclusion

This proposed multimodal driver stress detection framework highlights that carefully preprocessing biosignals and integrating them with deep temporal learning architectures leads to highly accurate, deployable systems for recognizing driver stress. CNN-BiLSTM-Attention sets a new benchmark of accuracy in offline evaluation, whereas CNN-GRU-Attention presents an efficient solution that can be practically adopted in real-world in-vehicle applications. Confirming strong generalization, the capability of this framework to deal with naturalistic, noisy physiological data supports the feasibility of AI-enabled in-vehicle driver stress monitoring for improving road safety and driver well-being in intelligent vehicles.

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